

Multivariable Limits

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I Introduction

Before we formally define a limit, we must define a function of several variables:

Definition. For a subset $D \subseteq \mathbb{R}^n$, a function $f(x_1, x_2, x_3, \dots, x_n)$ is a rule such that for each $(x_1, x_2, x_3, \dots, x_n)$ in D , $f(x_1, x_2, x_3, \dots, x_n)$ gives a definite number.

In other words, no matter how many variables a function has, it must give a definite number when you plug something in.

II Limits of Functions with Several Variables

Recall from calculus I that a limit exists if and only if the limit from the left and the limit from the right exist and are equal to each other. Consider a function $f(x, y)$. We want to find the limit of f as (x, y) approaches (x_0, y_0) , i.e.,

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$$

There are infinitely many ways to do this. So many that in general we say that if c is a smooth curve given by

$$x = x_1(t) \dots x_n = x_n(t) \text{ with } (x_1^0, \dots, x_n^0) = (x_1(t_0), \dots, x_n(t_0))$$

then the limit of $f(x_1, \dots, x_n)$ as $(x_1, \dots, x_n) \rightarrow (x_1^0, \dots, x_n^0)$ is defined by

$$\lim_{(x_1, \dots, x_n) \rightarrow (x_1^0, \dots, x_n^0)} f(x_1, \dots, x_n) = \lim_{t \rightarrow t_0} f(x_1(t), \dots, x_n(t))$$

Also note that a limit of a multivariable function exists if and only if for every smooth curve c passing through $(x_1^0, \dots, x_n^0)$

$$\lim_{(x_1, \dots, x_n) \rightarrow (x_1^0, \dots, x_n^0)} f(x_1, \dots, x_n) = L$$

[†]Definitions and example problems from MATH 22005: Analytic Geometry and Calculus III instructed by Dr. Gang Yu. Kent State University.

III Examples

Example 1. Find the limit of $f(x, y) = \frac{\sin(x+y)}{x-y}$ as $(x, y) \rightarrow (0, 0)$ along the i. x-axis, ii. y-axis, and iii. $y = 2x$.

Solution.

i. Making the y 's equal to zero (since we are solving along the x-axis),

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x+y)}{x-y} = \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

By L'Hospital's rule.

ii. Making the x 's equal to zero (since we are solving along the y-axis),

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x+y)}{x-y} = \lim_{y \rightarrow 0} \frac{\sin(y)}{-y} = -1$$

By L'Hospital's rule.

iii. Plugging in $y = 2x$,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x+y)}{x-y} = \lim_{x \rightarrow 0} \frac{\sin(x+2x)}{x-2x} = \lim_{x \rightarrow 0} \frac{\sin(3x)}{-x} = -3$$

By L'Hospital's rule. Notice how we got 3 different answers along different curves c . This implies that our limit does not exist, i.e.,

$$\lim_{\text{along } c_1} f(x_1, \dots, x_n) \neq \lim_{\text{along } c_2} f(x_1, \dots, x_n)$$